Isolating the Effect of Human Personality on LLM Response Quality

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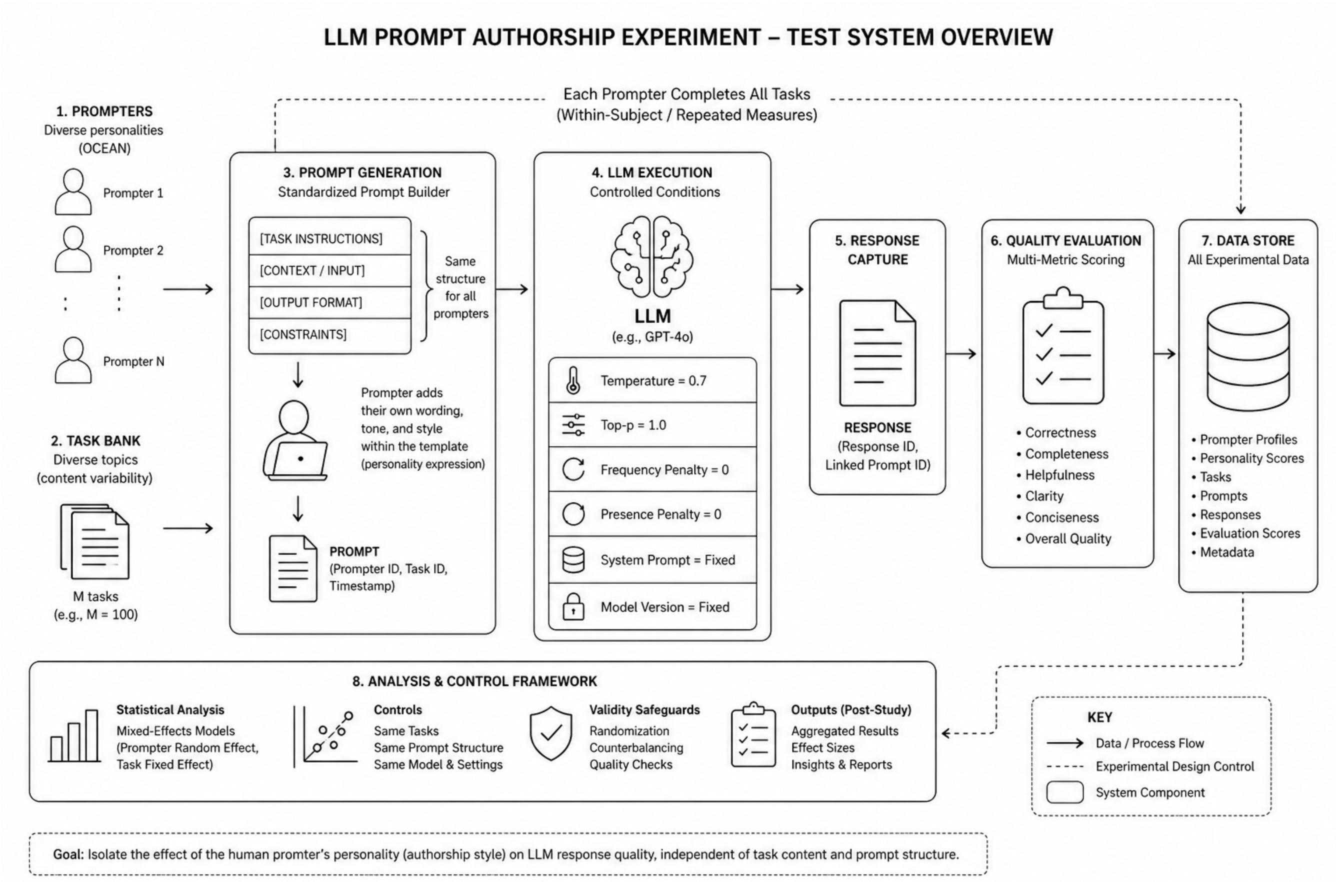
1. Abstract

This report documents the experimental design for testing whether the personality of a human LLM prompter affects LLM response quality independently of task content and prompt structure. The experiment is implemented in the LLM Prompt Eval v2 harness using prompt-authorship mode. In this mode, each experimental condition varies the authored request wording, task framing, implied priorities, and social stance toward the model, while protected task elements remain constant across conditions. The system preserves the task objective, contextual source material, output contract, and scope boundaries so that the measured differences can be attributed to the request-authorship variant rather than to changes in task substance or output requirements.

The experimental harness supports guided experiment construction, capability-aware model selection, YAML-backed compiled experiment artifacts, local execution through a runner, automatic scoring by a separate evaluator model, cached pairwise baseline comparisons, and human-readable result views grouped by personality type. The current report establishes the experimental rationale, validity controls, execution process, analysis plan, and future reporting structure.



2. Research Question and Test Objective

The central research question is: when task content and output structure are held constant, does the personality-driven authorship style of the prompt materially affect the quality of the LLM response?

The test objective is to isolate prompt authorship as the manipulated condition. The experiment is not designed to test whether one task is easier than another, whether one output contract is better than another, or whether a model can follow arbitrary stylistic overlays. It is designed to test whether different human prompting styles cause measurable differences in answer quality when the underlying work remains equivalent.

3. Experimental Hypothesis

Primary hypothesis: Personality-authored prompt variants will produce measurable differences in LLM response quality even when the task objective, context, output contract, and scope boundaries are preserved across conditions.

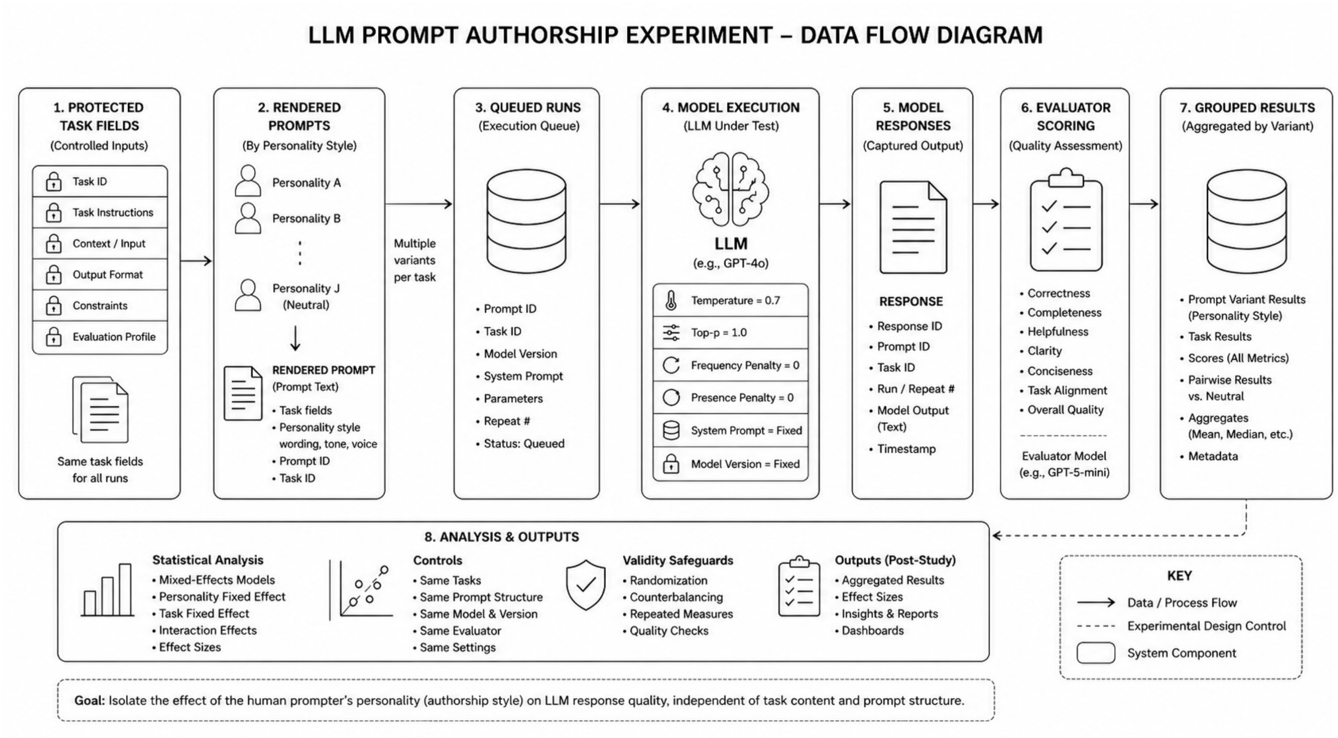
Null hypothesis: When the task objective, context, output contract, and scope boundaries are preserved, personality-authored request wording will not produce meaningful differences in response quality beyond normal model variance.

Secondary hypothesis: The magnitude and direction of quality differences will vary by task type. Some task types, such as recommendation, decision support, ideation, and risk analysis, may be more sensitive to prompt authorship because they depend heavily on framing, priorities, and tolerance for ambiguity. Other task types, such as constrained summarization, may show smaller authorship effects when the output contract is strict.

4. System Under Test

The LLM Prompt Eval v2 system is a local-first experiment harness for testing how prompt authorship affects LLM output quality. The current system state supports prompt-authorship experiments, guided drafting, model capability discovery, protected task fields, role-separated model execution, automatic scoring, cached pairwise baseline comparison, and result inspection.

Component	Test-engineering function
Review UI / Builder	Guided experiment drafting, model-role selection, protected field entry, rendered prompt preview, human approval before execution.
API service	Experiment builder endpoints, capability bootstrap, draft approval, execution metadata, result retrieval, and comparison surfaces.
Core package	Shared database models, schemas, settings, and validation logic.
Runner	Claims approved pending runs, executes model calls, captures outputs, and processes batches locally.
Model under test	The generation model that receives each rendered prompt and produces the response being evaluated.
Evaluator model	A separate model role used to score outputs and perform baseline comparisons.
Prompt transformer model	Role used during draft generation to support personality-authored request variants. Current support is restricted to OpenAI models.
Database	Stores experiment definitions, runs, raw outputs, scalar summaries, cached pairwise comparisons, and execution status.



5. Experimental Design

5.1 Independent Variable

The independent variable is prompt authorship style as represented by personality-authored request wording. In the current schema, each condition stores a prompt_authorship_variant and a rendered_prompt. The variant may change the request wording, the framing of the task, the implied priorities, and the social stance toward the model.

5.2 Controlled Variables

The design explicitly controls the task elements that would otherwise confound the experiment. These fields are protected and must remain constant across variants:

- Task objective
- Task context or source material
- Output contract
- Scope boundaries

The rendered prompt format enforces this separation by placing authored request wording above the protected task fields. This makes it possible to vary how the human asks while keeping what the human asks for constant.

5.3 Dependent Variables

The primary dependent variable is response quality as measured by automatic evaluator scoring. The system also supports pairwise baseline comparison, where each personality-authored output can be compared against the neutral or baseline output. Raw JSON is preserved for offline analysis and audit.

Measure	Purpose	Interpretation
Scalar quality score	Summarizes output quality using the evaluator model.	Used to compare quality across personality variants and task families.
Pairwise baseline comparison	Compares each variant against a baseline response.	Used to determine whether a variant wins, loses, or ties relative to neutral authorship.
Grouped results by personality type	Aggregates outcomes by modeled personality category.	Used to identify recurring performance patterns.
Top winners / biggest losers	Highlights high-impact positive and negative variants.	Used for executive-facing interpretation after sufficient data volume exists.
Raw response JSON	Preserves execution and scoring artifacts.	Used for audit, troubleshooting, and deeper offline analysis.

5.4 Personality Conditions

The current experiment model supports one neutral condition and nine named personality conditions. These conditions are generated using structured trait metadata rather than simple tone overlays.

Condition family	Conditions
Baseline	neutral
Personality-authored variants	Reformer, Helper, Achiever, Individualist, Investigator, Loyalist, Enthusiast, Challenger, Peacemaker
Trait metadata used for authorship	structure, directness, exploration, warmth, constraint, risk_focus

5.5 Rendered Prompt Structure

Every rendered prompt follows a fixed structure. This is a key validity control because it separates the manipulated authorship field from the protected task fields:

Request:
<personality-authored request wording>

Task objective:
<protected task objective>

Context:
<protected source/context material>

Output contract:
<protected output contract>

6. Validity Rationale

6.1 Internal Validity

The experiment is internally valid to the extent that observed output-quality differences can be attributed to the prompt-authorship condition rather than uncontrolled changes in the task. The design supports internal validity through protected content preservation, structured rendered prompt format, role separation, human approval, and validation checks that reject prompts if protected content is dropped or mutated.

Threat to validity	Design control	Residual risk
Task content drift	Task objective, context, and output contract are stored separately and preserved verbatim in rendered prompts.	Semantic drift may still occur if authored request wording implies priorities that conflict with the protected output contract.
Output contract inconsistency	The output contract is protected and included in every rendered prompt.	Models may still interpret priority cues differently across authored request variants.
Prompt mutation before execution	Rendered prompt preview and human approval are required before queuing.	Human reviewers must inspect the rendered prompt carefully.
Evaluator availability mismatch	Capability bootstrap marks evaluator eligibility only when the runner actually supports the model in evaluation mode.	Evaluator model behavior remains a source of measurement noise.
Run-state ambiguity	Experiment-level status rolls up queued, running, completed, failed, and completed_with_failures states.	Partial failures must be excluded or separately flagged in final analysis.

6.2 Construct Validity

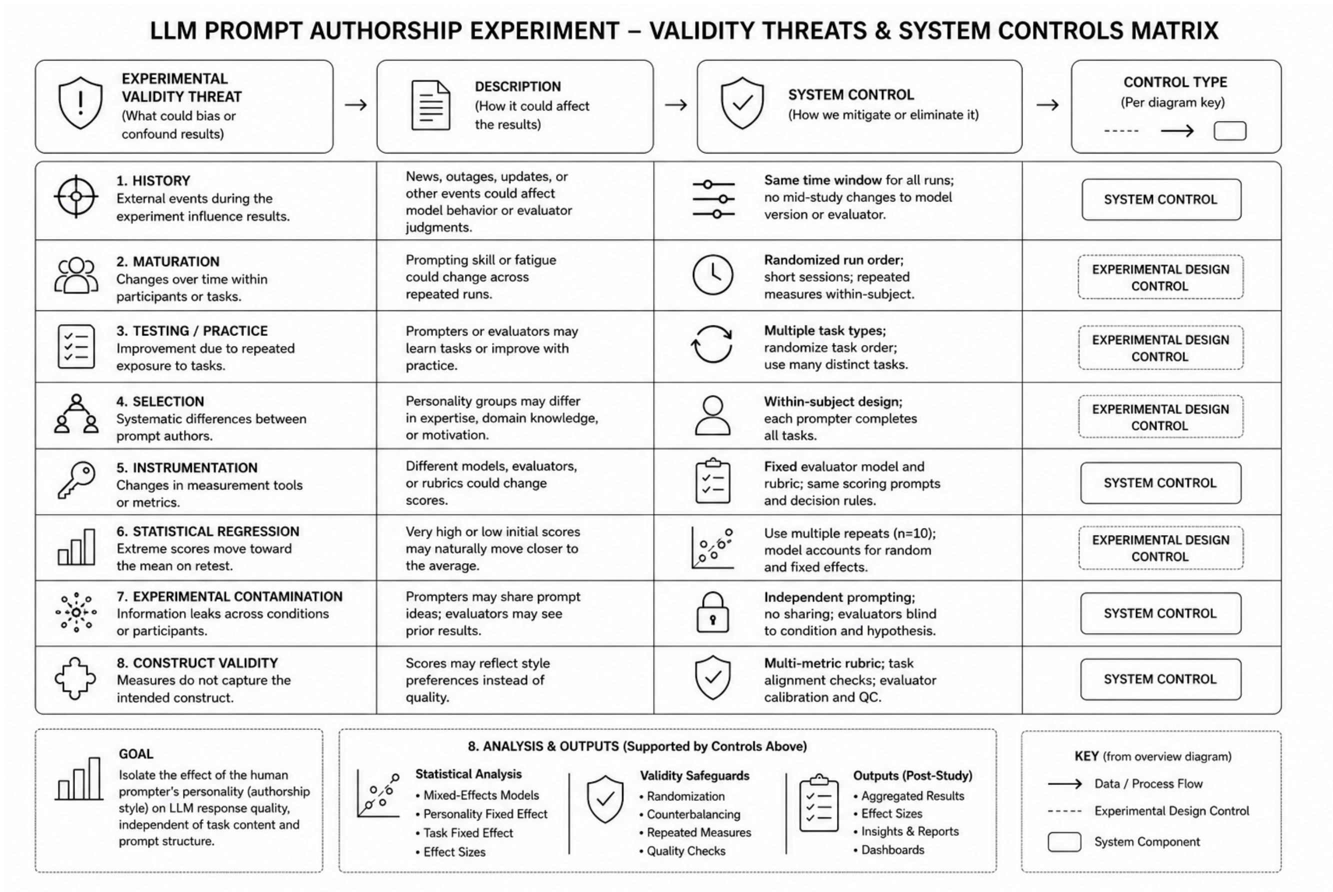
The construct being tested is not generic writing style. It is personality-influenced prompt authorship: the way a person frames a request, signals priorities, manages ambiguity, expresses constraints, and positions the relationship with the model. The design is stronger than a simple instruction such as “answer warmly” because the personality variation occurs in the authored request, not merely in an output-style directive.

6.3 External Validity

External validity depends on the breadth of task types, model families, and prompt variants included in the final run plan. The system currently supports OpenAI, xAI, and Ollama for generation and evaluation roles, subject to discovered capability and runner support. Results should be reported by model under test, evaluator model, task type, and personality condition before generalizing across broader LLM usage patterns.

6.4 Measurement Validity

The system uses an evaluator model to score outputs and to perform pairwise baseline comparisons. This enables repeatable, scalable scoring, but the final report should treat automated evaluation as a measurement instrument, not as ground truth. Stronger measurement validity can be achieved by using stable evaluator settings where available, reviewing sample outputs manually, and comparing scalar scores with pairwise baseline outcomes for consistency.



7. Execution Workflow

The operational workflow is designed to reduce accidental experiment contamination before runs are queued. The current workflow is:

1. Open the builder UI.

2. Bootstrap available model capabilities.

3. Enter experiment details, task objective, task context, output contract, personality types, and execution roles.

4. Generate the draft YAML.

5. Verify that the rendered prompt sample includes the full protected content.

6. Approve and queue the experiment.

7. Allow the runner to process queued runs.

8. Review run execution state, scalar summary scores, and pairwise baseline results.

8. Data Capture and Auditability

The experiment captures both human-readable results and raw execution artifacts. This supports both executive reporting and test-engineering audit. The database retains raw JSON for deeper offline analysis, while the UI provides grouped summaries and pairwise baseline comparisons for faster inspection.

Artifact	Why it matters
Compiled YAML artifact	Provides a durable, reviewable definition of the experiment before execution.
Rendered prompt	Shows the exact prompt sent to the model under test.
Run status and counts	Supports execution traceability and failure triage.
Model response	Primary observed output from the model under test.
Evaluator result	Primary automated quality measurement.
Pairwise baseline result	Relative comparison against baseline authorship.
Raw JSON	Audit trail for offline validation, debugging, and reproducibility checks.

9. Analysis Plan

After data collection, the analysis should be performed at three levels: aggregate performance, task-type sensitivity, and personality-condition effect. The primary comparison should be each personality-authored variant against the neutral baseline within the same task, same protected content, same output contract, same model under test, and same evaluator configuration.

Analysis view	Question answered	Recommended output
Overall personality lift/degradation	Do personality-authored prompts outperform, underperform, or match neutral prompts overall?	Mean score difference, median score difference, win/loss/tie count versus baseline.
Task-type sensitivity	Which task types are most affected by prompt authorship?	Grouped results by summarization, decision support, writing, explanation, recommendation, ideation, and risk analysis.
Personality-condition effect	Which personality-authored variants consistently improve or degrade outcomes?	Ranked personality table with confidence notes.
Model sensitivity	Does the effect vary by model under test?	Separate score and pairwise results by model family.
Evaluator consistency	Do scalar scores and pairwise comparisons agree?	Cross-check table comparing evaluator score deltas to pairwise outcomes.
Failure analysis	Which runs failed or produced unusable outputs?	Failed-run list and exclusion rationale.

10. Statistical and Engineering Interpretation Guidance

The final report should avoid overstating results from small sample sizes. The test should report both effect direction and effect magnitude. A personality variant that wins a few isolated examples should not be treated as generally superior unless the result is stable across tasks, models, and repeated runs. Conversely, a variant that performs poorly only on one task family may indicate a task-specific mismatch rather than a universal problem with that prompting style.

Where seeded deterministic behavior is unavailable, repeated runs are recommended for higher-confidence interpretation. The capability bootstrap identifies whether seed support exists for a provider/model combination. When seed support is missing, variance should be treated as part of the measurement environment and documented accordingly.

11. Known Limitations

- Some providers and models do not support seeded deterministic behavior, which may introduce run-to-run variance.
- Evaluator models are scalable measurement instruments but are not perfect substitutes for human expert review.
- Current semantic validation is stronger for verbatim preservation than for detecting subtle meaning drift.
- Older experiments created before protected-context and prompt-authorship fixes should be treated cautiously.
- Current offline reporting and export workflows are still evolving.
- Prompt-transformer support is currently restricted to OpenAI models.

12. Acceptance Criteria for a Valid Experiment Run

Criterion	Pass condition
Protected content preserved	Rendered prompt sample contains the complete task objective, task context, and output contract.
Condition isolation	Only request wording, framing, implied priorities, and social stance vary by personality condition.
Role selection verified	Model under test, evaluator model, and prompt transformer model are separately selected and capability-validated.
Human approval complete	Draft YAML and rendered prompt sample are reviewed before approval.
Execution complete	Experiment status is completed, or completed_with_failures with failures explicitly documented.
Scoring complete	Scalar score and pairwise baseline comparison are available for all included runs.
Audit artifacts retained	Rendered prompt, model output, evaluator result, and raw JSON remain available for inspection.

13. Findings Summary

13.1 Dataset Analyzed

The findings summarized in this section are based on experiments 174 through 194. The analyzed set includes 21 completed experiments across seven task types, with three experiments per task type. Each experiment includes one neutral baseline condition and nine personality-authored conditions. Each condition was executed using repeated runs, and each completed run received a scalar quality score from the evaluator model.

Dataset element	Count / scope
Experiments analyzed	21
Experiment ID range	174–194
Task types	7
Experiments per task type	3
Personality-authored variants	9
Neutral baseline	Included for reference and pairwise comparison
Primary measurement	Scalar evaluator score
Secondary validation	Pairwise comparison versus neutral
Maximum scalar score	10

The scalar evaluator score was used as the primary measurement instrument because it provides the most precise basis for estimating quality differences and recoverable lift. Pairwise comparison results were used as a directional validation layer, not as the primary measure of effect size.

13.2 Primary Finding: Personality-Driven Prompt Style Affects Output Quality

The scalar results show measurable variation in response quality by personality-authored prompt style. The effect was not uniform across all tasks or all personality conditions, but the difference between stronger and weaker prompt styles was large enough to be operationally meaningful.

Across the 21 completed experiments, the average gap between the strongest and weakest personality-authored style was 1.50 points on a 10-point scale. Expressed as recoverable improvement from the weaker style, the average lift was 28.3%. The observed range was 11.6% to 72.8%.

Primary scalar finding	Result
Average best-versus-worst score delta	1.50 points
Average recoverable lift	28.3%
Minimum observed recoverable lift	11.6%
Maximum observed recoverable lift	72.8%
95% confidence interval for average recoverable lift	21.0% to 35.6%

The confidence interval indicates that, for this experiment set, the expected average recoverable lift is unlikely to be a small or negligible effect. Even the lower bound of the interval remains above 20%, which supports the conclusion that prompt-authorship alignment has a material effect on output quality.

The variation is also meaningful at the score level. A 1.50-point average spread on a 10-point evaluator scale is large enough to affect practical interpretation of output quality. In production terms, this is the difference between an answer that may be merely acceptable and one that is materially more useful, complete, or aligned with the task.

13.3 Recoverable Lift Analysis

The most relevant measure for product interpretation is recoverable lift. Recoverable lift estimates how much improvement may be obtained when a weaker personality-authored prompt style is transformed into the strongest observed task-aligned style for the same task.

The calculation used in this analysis was:

Measure	**Calculation**
Score delta	Best personality-authored score minus worst personality-authored score
Recoverable lift	Score delta divided by the worst personality-authored score

The denominator is the weaker starting score because the product question is improvement from the user's current prompt style. A user does not start from the maximum score. The user starts from the quality level produced by the prompt they actually wrote. Therefore, the practical improvement estimate is the score gap divided by that starting score.

For example, in one recommendation experiment, the strongest personality-authored style scored 7.70 and the weakest scored 4.45. The score delta was 3.24 points, producing a recoverable lift of 72.8%. In a more constrained summarization experiment, the strongest score was 8.28 and the weakest score was 7.41. The score delta was 0.86 points, producing a recoverable lift of 11.6%.

Example	Task type	Best score	Worst score	Delta	Recoverable lift
High-lift example	Recommendation	7.70	4.45	3.24	72.8%
Low-lift example	Summarization	8.28	7.41	0.86	11.6%

These examples illustrate the main interpretation of the data. The value is not in declaring one personality universally better than another. The value is in identifying the degree to which task-specific prompt alignment can recover quality that is otherwise lost when a natural prompting style is poorly matched to the work being requested.

13.4 Variation by Task Type

Recoverable lift varied substantially by task type. The strongest lift appeared in task categories that require judgment, prioritization, tradeoff analysis, recommendation, or exploration. The smallest lift appeared in more bounded tasks where the output target is more constrained.

Task type	Avg. best-worst delta	Avg. recoverable lift	95% confidence interval for lift	Interpretation
Decision support	1.91	47.5%	6.1% to 88.9%	High effect, high variation

Task type	Avg. best-worst delta	Avg. recoverable lift	95% confidence interval for lift	Interpretation
Recommendation	1.94	38.1%	Directional only; high spread	High effect, low precision
Ideation	1.63	29.4%	5.8% to 52.9%	High effect, moderate variation
Risk analysis	1.49	28.7%	26.3% to 31.2%	High effect, high consistency
Explanation	1.26	19.6%	10.8% to 28.4%	Moderate effect
Writing	1.15	19.4%	9.9% to 28.9%	Moderate effect
Summarization	1.10	15.4%	6.8% to 24.1%	Lower but measurable effect

Because each task type is represented by three experiments, the task-level confidence intervals should be interpreted as directional rather than final. The strongest precision appears in risk analysis, where the average recoverable lift was 28.7% and the observed variation across the three experiments was low. Recommendation showed a high average lift but also high experiment-to-experiment variation, so the directional conclusion is strong, but the exact lift estimate should be treated with caution.

The task-level pattern is still clear. Decision support, recommendation, ideation, and risk analysis produced the largest recoverable lift. These tasks require the model to infer priorities, evaluate alternatives, handle ambiguity, and decide how to frame the answer. That makes them more sensitive to the authorship style of the prompt.

Summarization produced the lowest average recoverable lift, but the effect was still measurable. This is consistent with the structure of the task. Summarization constrains the model more tightly because the answer is anchored to source material and preservation of meaning. There is less room for prompt-authorship style to change the result.

13.5 Variation by Personality Type

Personality-level results were analyzed in two ways. The first view examined average scalar score by personality-authored style across all 21 experiments. The second view examined average recoverable lift to the best observed task-aligned style. The second view is more important for the product interpretation because it estimates how much improvement could be recovered for a user whose natural prompting style is not well matched to the task.

Personality style	Avg. scalar score	95% confidence interval	Interpretation
Enthusiast	6.91	6.56 to 7.25	Highest average score; lowest recoverable gap
Peacemaker	6.76	6.44 to 7.07	Strong average performance
Individualist	6.74	6.28 to 7.20	Strong average performance, higher variation
Loyalist	6.68	6.34 to 7.02	Mid-range, relatively stable
Achiever	6.63	6.33 to 6.94	Mid-range, relatively stable
Reformer	6.58	6.15 to 7.01	Mid-range, higher variation
Helper	6.57	6.18 to 6.97	Mid-range, higher variation
Investigator	6.39	6.02 to 6.77	Lower average score
Challenger	5.88	5.37 to 6.39	Lowest average score; highest recoverable gap

This table should not be read as a universal ranking of personality value. It is an aggregate view across multiple task types. A personality style that performs poorly in one task category may be more useful in another. However, the aggregate result does show that some authorship styles are more frequently aligned with evaluator-preferred outputs in this experiment set.

The recoverable-lift view shows a clearer implication.

Personality style	Avg. point gap to best task-aligned style	Avg. recoverable lift to best task-aligned style	95% confidence interval for lift
Challenger	1.37	26.0%	18.1% to 34.0%
Investigator	0.85	14.0%	10.5% to 17.5%
Reformer	0.67	11.0%	7.4% to 14.5%
Helper	0.67	10.9%	7.2% to 14.6%
Achiever	0.61	9.6%	5.9% to 13.2%
Loyalist	0.57	8.9%	6.1% to 11.6%
Individualist	0.50	8.5%	4.4% to 12.6%
Peacemaker	0.49	7.4%	4.9% to 9.8%

Personality style	Avg. point gap to best task-aligned style	Avg. recoverable lift to best task-aligned style	95% confidence interval for lift
Enthusiast	0.34	5.1%	3.2% to 7.1%

This view indicates that the largest recoverable opportunity appears where the natural authorship style is most often misaligned with the task. In this data set, the Challenger style produced the largest average gap to the best task-aligned style. That does not mean Challenger users are less capable. It means that the Challenger-authored prompt formulation was often less aligned with the response characteristics rewarded by the evaluator for these tasks. This is exactly the type of pattern a prompt transformation layer is intended to correct.

The confidence intervals also show that the personality-level estimates are more stable than the task-level estimates because each personality condition appears across 21 experiments. The intervals are still not narrow enough to support universal personality claims, but they are sufficient to show that the recoverable-lift pattern is not driven by a single isolated task.

13.6 Scalar Score and Pairwise Comparison Interpretation

The scalar score was the primary basis for the findings because it provides the most precise measurement of output quality. All score deltas, recoverable-lift calculations, task-level averages, personality-level averages, and confidence intervals in this section were derived from scalar evaluator scores.

Pairwise baseline comparisons were used only as directional validation. The pairwise comparison asks whether a personality-authored output was better than, worse than, or tied with the neutral baseline. That is useful, but it is not the same question as the recoverable-lift analysis. Recoverable lift compares the weakest and strongest personality-authored styles within the same task context. Pairwise comparison compares each condition against neutral.

Measurement view	Question answered	Use in this section
Scalar evaluator score	How much did output quality change?	Primary measure
Best-worst scalar spread	How much lift is recoverable through task-aligned transformation?	Primary product interpretation
Pairwise comparison versus neutral	Does the scalar direction generally align with direct preference against baseline?	Directional validation only

The pairwise results generally supported the scalar findings where the scalar differences were large. In particular, weak scalar performers tended to show weak pairwise performance against neutral. However, pairwise results were not used to calculate effect size because they do not provide the same precision as scalar scoring and do not directly compare every personality-authored condition to every other personality-authored condition.

The appropriate interpretation is therefore as follows: the scalar data establishes the measured effect; pairwise comparison provides a secondary reasonableness check. Where scalar score and pairwise direction agree, confidence in the pattern is stronger. Where they do not fully agree, the scalar result remains the primary measurement, but the finding should be treated as less mature until additional data or human review is available.

13.7 Statistical Variation and Confidence

The overall average recoverable lift across the 21 experiments was 28.3%, with a standard deviation of 15.9 percentage points. The estimated 95% confidence interval for the average recoverable lift was 21.0% to 35.6%. This means the central effect is large enough that the overall finding is unlikely to depend on a small number of unusual experiments.

Statistical measure	Result
Mean recoverable lift	28.3%
Standard deviation	15.9 percentage points
Standard error	3.5 percentage points
95% confidence interval	21.0% to 35.6%
Number of experiments	21

The task-type confidence calculations are less precise because each task type has three experiments. For that reason, task-level intervals should be treated as a guide to relative sensitivity, not as final population estimates. The strongest task-level confidence is present where the mean effect is large and the variation is low. Risk analysis is the clearest example. Recommendation is the opposite case: the mean effect is large, but variation across the three experiments is also large, so the exact estimate is less precise.

The personality-level confidence calculations are more stable because each personality-authored condition is represented across all 21 experiments. At the personality level, the confidence intervals support the conclusion that some authorship styles have larger recoverable gaps than others. The most important product implication is not that one style is universally best. It is that some styles create a larger average opportunity for task-aligned transformation.

For reporting purposes, the confidence level should be stated carefully. The overall effect has moderate confidence. The personality-level recoverable-lift estimates have moderate directional confidence. The task-level ranking has directional confidence, with stronger confidence for consistent categories such as risk analysis and weaker precision for high-variation categories such as recommendation.

13.8 Data-Quality Notes and Exclusions

The neutral baseline was retained for baseline interpretation and pairwise directional validation but excluded from the best-versus-worst personality-authored spread. This keeps the recoverable-lift analysis aligned to the question of transformation between personality-authored prompt styles.

The current findings should be interpreted as an initial empirical readout. The experiment set is sufficient to show a meaningful and measurable authorship effect under controlled conditions. Additional experiments would improve precision, especially at the task-type level, where each task category currently has three experiments.

The evaluator model remains a measurement instrument, not ground truth. The scalar scores are useful because they are consistent, repeatable, and precise enough for effect-size estimation. However, they should still be interpreted with the known limitations of automated evaluation. Human review of representative outputs would strengthen the final interpretation, especially for task categories with large variation or mixed scalar-versus-pairwise signals.

14. Executive Summary

The results from experiments 174 through 194 provide empirical support for the central hypothesis of this study: personality-driven prompt authorship affects LLM response quality when task content, output requirements, model role, evaluator role, and execution structure are controlled.

The most important finding is not that one personality type is universally superior. The more useful finding is that different prompting styles create different levels of task alignment. When a user's natural prompt style is poorly aligned to the task, the model may produce a lower-quality response even though the underlying request is valid. When that same request is transformed into a better-aligned task style, the response quality can improve materially.

Across the completed experiment set, the average gap between the strongest and weakest personality-authored prompt style was **1.50 points on a 10-point scale**. Expressed as recoverable improvement from the weaker starting style, the average lift was **28.3%**. The observed improvement range was approximately **12% to 73%**, depending on task type and starting prompt style.

Executive finding	Result
Experiments analyzed	21
Task types represented	7
Average best-versus-worst scalar delta	1.50 points
Average recoverable lift	28.3%
Maximum recoverable lift	72.8%
Observed recoverable lift range	11.6% to 72.8%
95% confidence interval for average recoverable lift	21.0% to 35.6%

The effect was not evenly distributed across task types. The largest recoverable lift appeared in tasks that require judgment, prioritization, tradeoff analysis, recommendation, and ambiguity management. Decision support, recommendation, ideation, and risk analysis showed the highest average lift. More bounded tasks, such as summarization and explanation, still showed measurable improvement, but with smaller effect sizes.

Task type	Average recoverable lift
Decision support	47.5%
Recommendation	38.1%
Ideation	29.4%
Risk analysis	28.7%
Explanation	19.6%
Writing	19.4%
Summarization	15.4%

The personality-level results also support the prompt-transformation thesis. The data shows that some natural authorship styles have larger recoverable gaps than others. In this experiment set, the largest average recoverable opportunity appeared where the source prompt style was most often misaligned with the task. This should not be interpreted as a judgment about the person or their capability. It should be interpreted as evidence that LLMs respond differently to different human communication patterns, and that prompt transformation can reduce the penalty created by that mismatch.

The pairwise comparisons were used as directional validation only. The primary findings were derived from scalar evaluator scores because scalar scores provide the precise measurement needed to calculate effect size and recoverable lift. Pairwise comparisons were useful as a reasonableness check, especially when weak scalar performers also underperformed against neutral baseline. However, pairwise comparisons were not used as the primary basis for the improvement estimates because they compare each variant to neutral, not every personality-authored style to every other style.

The practical implication is clear: prompt quality is not only a function of the user's intent or the model's capability. It is also affected by the way the user frames the request. This creates a product opportunity for a translation layer that preserves the user's intent while rendering the prompt in a style better aligned to the task, model, and desired output.

The results therefore support the thesis: measurable value can be created by helping users convert natural human prompting behavior into task-aligned LLM instructions. The value is not personality labeling for its own sake. The value is recoverable output quality.

16. Conclusion

The LLM Prompt Eval v2 harness provides a credible experimental structure for isolating the effect of human personality-driven prompt authorship on LLM response quality. The key engineering strength is the separation of mutable authored request wording from protected task content. This enables a cleaner causal interpretation than experiments that change the whole prompt or merely instruct the model to answer in a different tone.

The design is valid for local experimentation when protected content preservation is verified, model roles are capability-validated, failed runs are handled explicitly, and automated evaluation is interpreted as a measurement instrument with known limitations. After data collection, the report can be completed by adding empirical findings, charts, representative examples, and an executive summary that translates the results into product and go-to-market implications.